

Case Study and Requirements

A trading floor support center employs 765 staff. They have recently expanded and, as a result, need to move to a new building. A building has been identified but has no network. This means that before they can move out, a new network service needs to be designed and implemented in the new building. The existing network comprises the following elements: The new building is expected to have four floors with two departments on each floor, for example:

1. First floor - (Sales and Marketing Department - 150 users expected, Human Resources and Logistics Department - 150 users expected).
2. Second floor - (Finance and Accounts Department - 150 users expected, Administrator and Public Relations Department - 150 users expected).
3. Third floor - (ICT - 150 users expected, Server Room - 15 devices expected).
4. Fourth floor – (Executive Floor - 15 users expected)

Therefore, as a key member of the Networks Team, you have been tasked with designing a network for the new building. At this stage, a logical design is required, which shows the measures you would put in place to ensure that the new network meets the current business need and is future-proofed:

- Use Cisco Packet Tracer to design and implement the network solution
Use a hierarchical model providing redundancy at every layer; i.e., two routers and two multilayer switches are expected to be used to provide redundancy.
- The network is also expected to connect to at least two ISPs to provide redundancy, and each router to the switch connected to the two ISPs.

- Each department is required to have a wireless network for its users.
- Each department should be in a different VLAN and a different subnet.
- Provided with a base network of 172.16.1.0, carry out subnetting to allocate the correct number of IP addresses to each department.
- The company network is connected to the static, public IP addresses (Internet Protocol) 195.136.17.0/30, 195.136.17.4/30, 195.136.17.8/30, and 195.136.17.12/30, which are connected to the two Internet providers
- Configure basic device settings such as hostnames, console password, enable password, banner messages, and disable IP domain lookup.
- Devices in all departments are required to communicate with each other using the respective multilayer switch configured for inter-VLAN routing.
- The multilayer switches are expected to perform both routing and switching functionalities; thus, IP addresses will be assigned.
- All devices in the network are expected to obtain an IP address dynamically from the dedicated DHCP servers located at the server room.
- Devices in the server room are to be allocated an IP address statically.
- Use OSPF as the routing protocol to advertise routes on both the routers and multilayer switches

- Configure SSH on all routers and layer three switches for remote login.
- Configure port security for the Finance and Accounts department to allow only one device to connect to a switchport. Use sticky methods to obtain the MAC address and enable violation mode shutdown.
- Configure PAT to use the respective outbound router interface IPv4 address and implement the necessary ACL rules.
- Test communication to ensure everything configured is working as expected

1. Basic settings for all devices + SSH on routers and L3 switches + device management (hostname, passwords, banners).
2. VLAN assignment + all access and trunk ports on L2 and L3 switches (Enterprise VLAN segmentation).
3. Switchport security for sensitive departments (Finance) + Port Security + BPDU Guard + PortFast.
4. Subnetting and IP addressing (VLSM design for all VLANs + management + server networks).
5. OSPF on routers and L3 switches (single/multi-area + passive interfaces + route optimization).
6. Static IP addressing for server room devices (DNS, Web, File, DHCP, Syslog, NTP servers).
7. DHCP server configuration (centralized DHCP + DHCP pools + excluded addresses).
8. Inter-VLAN routing on L3 switches + IP helper-address for DHCP relay.
9. Wireless network configuration (multiple SSIDs: Corporate + Guest + VLAN separation + WPA2/WPA3).
10. PAT (NAT Overload) + Access Control Lists (ACLs) for traffic control between VLANs + Internet access rules.
11. Redundancy configuration (HSRP/VRRP for gateway redundancy + EtherChannel for link redundancy + STP optimization).
12. Network security enhancements (DHCP Snooping + Dynamic ARP Inspection + management VLAN isolation).
13. Network services configuration (DNS + NTP + Syslog + optional SNMP monitoring).
14. Testing and verification of all configurations (routing, VLANs, DHCP, NAT, OSPF, ACLs, connectivity tests).

Project Status, Progress & Learning Scope

This project is a self-directed learning project developed independently to strengthen my practical networking skills and apply the networking concepts I have studied using Cisco Packet Tracer.

The project is currently in progress and has not yet been fully completed**. The current version includes the network design, VLSM subnetting, VLAN segmentation, switch configuration, and basic network security configurations that I have studied and practiced so far.

Completed Work

The following components have been designed, configured, and practiced as part of the current project:

- * Designed the network using a hierarchical network design approach.
- * Configured device hostnames.
- * Configured `enable secret` for privileged EXEC mode.
- * Configured console line password protection using `line console 0`.
- * Configured an MOTD security banner.
- * Disabled DNS lookup using `no ip domain-lookup`.
- * Created and assigned VLANs.
- * Configured access ports and trunk ports across Layer 2 and Layer 3 switches.
- * Implemented VLAN segmentation across the organization.
- * Configured Port Security for the Finance and Accounting department to allow only one device per designated switch port.
- * Performed VLSM subnetting to allocate appropriate IP address ranges according to the requirements of each department, including addressing for ISP router connections.

Not Yet Implemented / Areas Requiring Further Study

The following components have **not yet been implemented in the current version of the project**. They represent the next stages of my learning and require further study and practical application before being incorporated into the project:

- * OSPF dynamic routing.
- * Static IP addressing for the server room.
- * DHCP server configuration.
- * Wireless network configuration.
- * NAT/PAT, ACLs, and Internet access control policies.
- * Network redundancy and high-availability mechanisms, including HSRP and VRRP.
- * Additional network security mechanisms, including Dynamic ARP Inspection and DHCP Snooping.
- * Management VLAN isolation and additional management-plane security.
- * Additional network services.
- * Final configuration verification, connectivity testing, and troubleshooting.

Reason for the Current Project Status

The project remains incomplete because I am developing it progressively as part of my ****self-learning process**** rather than implementing technologies that I have not yet studied and practiced sufficiently.

In addition, the current Packet Tracer project file has reached a saving limitation, which prevented me from continuing to save further changes to the same project file.

Therefore, the remaining configurations have intentionally been identified as ****areas for further study and practical implementation****, rather than being presented as completed work.

****Project Status: In Progress — Self-Directed Learning Project****